

--1) Retrieve all books in the ' Non - Fiction ' genre .

ANSWER:

```
SELECT * FROM books
WHERE genre = 'Non-Fiction';
```

--2)Find books published after the year 1950.

ANSWER:

```
SELECT * FROM books
WHERE published_year > 1950;
```

--3) Select all customers from Equatorial Guinea.

ANSWER:

```
SELECT * FROM customers
WHERE COUNTRY = 'Equatorial Guinea';
```

--4)Show orders placed in august 2023.

ANSWER:

```
SELECT * FROM orders
WHERE Order_Date BETWEEN '2023-08-01' AND '2023-08-31';
```

--5) Select the total stock of books available.

ANSWER:

```
SELECT SUM(Stock) AS Stock_Available
FROM books;
```

--6) Find the details of the most expensive book.

ANSWER:

```
SELECT * FROM books
ORDER BY Price DESC
LIMIT 1;
```

--7)Show all customers who ordered more than 1 quantity of a book.

ANSWER:

```
SELECT * FROM orders  
WHERE Quantity >1;
```

--8)Retrieve all books where the total amount exceeds \$20.

ANSWER:

```
SELECT * FROM orders  
WHERE Total_Amount > 20;
```

--9)List all the genres available in the books table.

ANSWER:

```
SELECT DISTINCT Genre FROM books;
```

--10)Find the book with the lowest stock:

ANSWER:

```
SELECT * FROM books  
ORDER BY Stock ASC  
LIMIT 1;
```

--11) Calculate the total revenue generated .

ANSWER:

```
SELECT SUM(Total_Amount) AS revenue_generated  
FROM orders;
```

--Advance questions:

--1)Retrieve the total amount of books sold for each genre.

ANSWER:

```
SELECT b.genre , SUM(o.quantity) AS Total_Books_Sold
```

```
FROM orders o  
JOIN books b ON b.Book_ID = o.Book_ID  
GROUP BY b.genre;
```

--2) Find the average price of books in the "Fantasy" genre.

ANSWER:

```
SELECT AVG(Price) AS Avg_Price  
FROM books  
WHERE genre = 'Fantasy';
```

--3) List customers who have placed atleast 2 orders.

ANSWER:

```
SELECT Customer_ID, COUNT(Order_ID) AS ORDER_COUNT  
FROM orders  
GROUP BY(Customer_ID)  
HAVING COUNT(Order_ID) >=2;
```

--4) Find the most frequently ordered book.

ANSWER:

```
SELECT Book_ID, COUNT(Order_ID) AS Order_Count  
FROM orders  
GROUP BY Book_ID  
ORDER BY Order_Count DESC LIMIT 1;
```

--5) Show the two most expensive books by 'Fantasy' genre.

ANSWER:

```
SELECT * FROM books  
WHERE Genre = 'Fantasy'  
ORDER BY Price DESC  
LIMIT 2;
```

--6) Retrieve the total quantity of books sold by each author.

ANSWER:

```
SELECT b.Author, SUM(o.Quantity) AS Total_Books_Sold
FROM orders o
JOIN books b ON b.Book_ID = o.Book_ID
GROUP BY b.Author;
```

--7) List of cities where customers who spent over \$30 are located.

ANSWER:

```
SELECT DISTINCT c.City
FROM orders o
JOIN customers c ON o.Customer_ID = c.Customer_ID
WHERE o.Total_Amount > 30;
```

--8) Find the customer who spent the most on orders .

ANSWER:

```
SELECT c.Customer_ID, c.Name, SUM(o.Total_Amount) AS Total_Spent
FROM orders o
JOIN customers c ON o.Customer_ID = c.Customer_ID
GROUP BY c.Customer_ID, c.Name
ORDER BY Total_Spent DESC LIMIT 1;
```